

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.

(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

Question 1

In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.

Rubric-Based Answer

Problem Understanding (20%)

Inputs: total data received = 8000 bits, time = 4 seconds, packets sent = 200, packets received = 180. Outputs: throughput, packet-loss percentage, network classification and complete summary. Constraints: time and packets sent must not be zero.

Algorithm Design (30%)

The algorithm validates inputs first, calculates throughput as data/time , calculates packet loss as $((\text{sent}-\text{received})/\text{sent})*100$, applies both classification conditions, and prints a complete performance summary.

Use of Control Structures (20%)

Uses IF/ELSE validation and classification conditions. Validation prevents division-by-zero errors.

Pseudocode Structure (10%)

Uses clear START, INPUT, IF, CALCULATE, DISPLAY and END steps in a consistent sequence.

Completeness (20%)

Handles invalid time, zero packets sent, valid calculation, classification and summary output.

Pseudocode

START

SET dataReceived = 8000

SET time = 4

SET packetsSent = 200

SET packetsReceived = 180

IF time <= 0 THEN

DISPLAY "Error: Time must be greater than zero."

STOP

END IF

IF packetsSent <= 0 THEN

DISPLAY "Error: Packets sent must be greater than zero."

STOP

END IF

throughput = dataReceived / time

lostPackets = packetsSent - packetsReceived

packetLoss = (lostPackets / packetsSent) * 100

IF throughput > 1500 AND packetLoss < 10 THEN

status = "Efficient"

ELSE

status = "Inefficient"

END IF

DISPLAY "NETWORK PERFORMANCE SUMMARY"

DISPLAY "Throughput:", throughput, "bps"

DISPLAY "Packets Sent:", packetsSent

DISPLAY "Packets Received:", packetsReceived

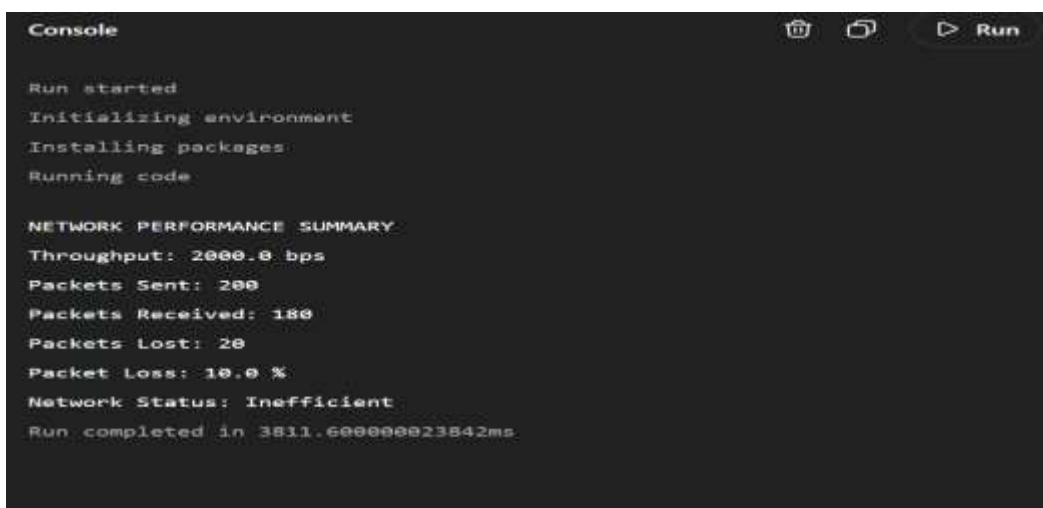
DISPLAY "Packets Lost:", lostPackets

DISPLAY "Packet Loss:", packetLoss, "%"

DISPLAY "Network Status:", status

END

Expected Output



```
Console
Run started
Initializing environment
Installing packages
Running code

NETWORK PERFORMANCE SUMMARY
Throughput: 2000.0 bps
Packets Sent: 200
Packets Received: 180
Packets Lost: 20
Packet Loss: 10.0 %
Network Status: Inefficient
Run completed in 3811.600000023842ms
```

Reason: The throughput is greater than 1500 bps, but packet loss is not less than 10%.

Question 2

An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

Rubric-Based Answer

Problem Understanding (20%)

Inputs are network links and their bandwidth utilization values. Outputs are stored utilization values, overload status and backup-link recommendation. Constraints include continuous monitoring and an overload threshold of greater than 80%.

Algorithm Design (30%)

The algorithm repeatedly reads utilization for every link, stores it in a list, checks the 80% threshold, and recommends backup traffic switching when required.

Use of Control Structures (20%)

Uses an outer monitoring loop and an inner loop over all links, plus IF/ELSE conditions for overload detection.

Pseudocode Structure (10%)

Uses clear initialization, collection, storage, checking, recommendation and waiting steps.

Completeness (20%)

Handles all links repeatedly and provides action for both normal and overloaded conditions.

Pseudocode

START

SET links = ["Link1", "Link2", "Link3", "Link4"]

SET utilization = empty list

SET monitoringInterval = 60 seconds

WHILE TRUE DO

CLEAR utilization

FOR each link IN links DO

usage = GET_BANDWIDTH_UTILIZATION(link)

APPEND usage TO utilization

DISPLAY link, "Utilization:", usage, "%"

IF usage > 80 THEN

DISPLAY "WARNING:", link, "is overloaded."

DISPLAY "Recommendation: Switch traffic to backup link."

ELSE

DISPLAY link, "is operating within limit."

END IF

END FOR

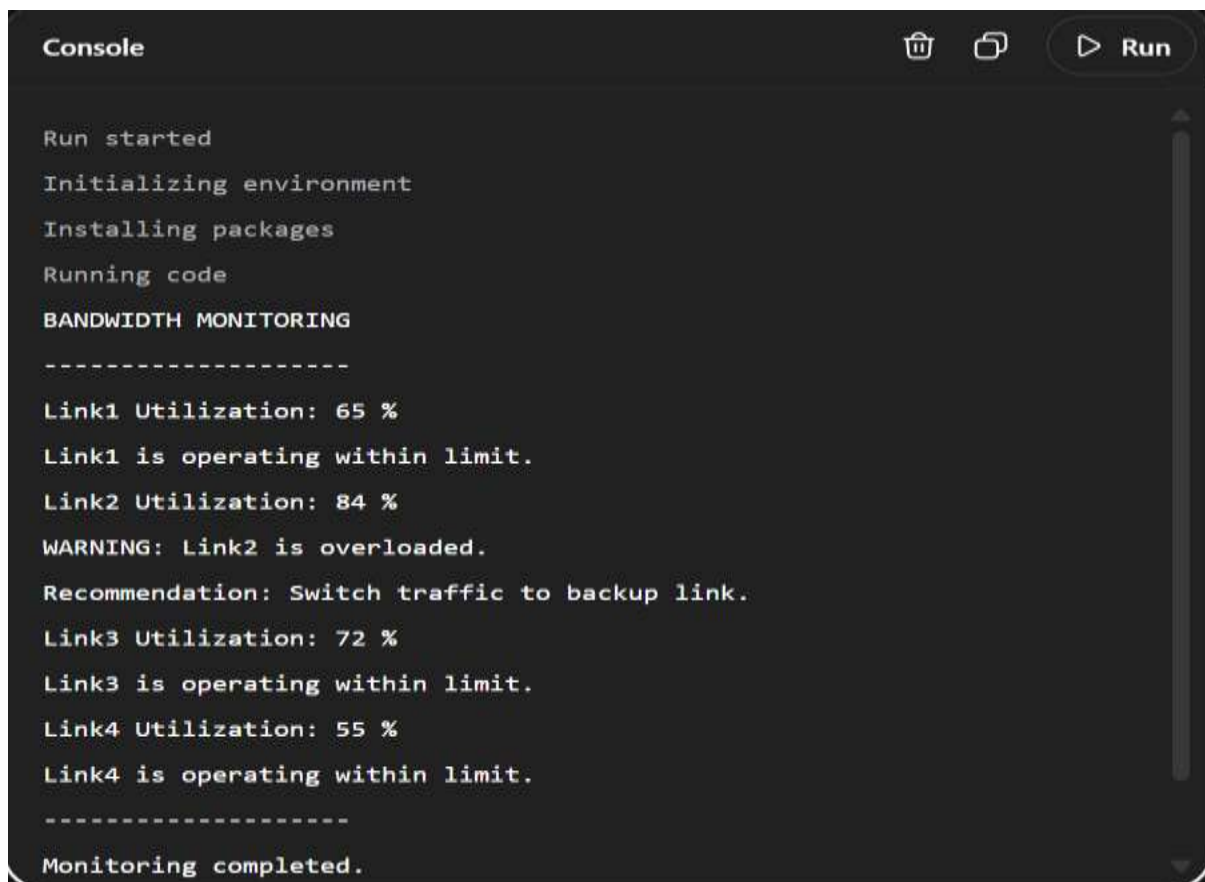
***DISPLAY "Monitoring completed. Next check after",
monitoringInterval, "seconds."***

WAIT monitoringInterval

END WHILE

END

Output:



```
Console
Run started
Initializing environment
Installing packages
Running code
BANDWIDTH MONITORING
-----
Link1 Utilization: 65 %
Link1 is operating within limit.
Link2 Utilization: 84 %
WARNING: Link2 is overloaded.
Recommendation: Switch traffic to backup link.
Link3 Utilization: 72 %
Link3 is operating within limit.
Link4 Utilization: 55 %
Link4 is operating within limit.
-----
Monitoring completed.
```

Explanation

The algorithm stores each link's utilization in an array/list and repeatedly checks all links. If utilization exceeds 80%, it raises a warning and recommends shifting traffic to a backup link. This reduces the chance of congestion, improves link utilization and supports continuous bandwidth management.

Question 3

A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

Rubric-Based Answer

Problem Understanding (20%)

Inputs are bandwidth, delay and utilization for every link/path connecting headquarters to six branches. Outputs are link-quality scores, best path, alternate path and administrator alerts. Constraints include changing values, periodic monitoring, overload/failure detection and continuous adaptation.

Algorithm Design (30%)

The algorithm collects metrics, stores them, calculates a normalized quality score, selects the highest-scoring valid path, checks overload/failure, and switches to an alternate path when required.

Use of Control Structures (20%)

Uses an outer periodic loop, loops through links/paths, IF conditions for validity and overload, and a selection process for the maximum score.

Pseudocode Structure (10%)

Clearly separates data collection, score calculation, path selection, failure handling, alerting and waiting.

Completeness (20%)

Handles changing metrics, six branches, multiple paths, overloaded links, failed links, alternate selection and alerts.

Link Quality Formula

One suitable normalized score is:

$$\text{Quality Score} = 0.5 \times \text{Normalized Bandwidth} + 0.3 \times (1 - \text{Normalized Delay}) + 0.2 \times (1$$

– Normalized Utilization)

Higher score = better overall link quality. Failed links are assigned a score of -1 and overloaded links can be excluded from primary-path selection.

Pseudocode

START

SET branches = ["B1", "B2", "B3", "B4", "B5", "B6"]

SET monitoringInterval = 60 seconds

WHILE TRUE DO

FOR each branch IN branches DO

paths = GET_ALL_PATHS("HQ", branch)

bestPath = NONE

bestScore = -1

alternatePath = NONE

alternateScore = -1

FOR each path IN paths DO

bandwidth = GET_BANDWIDTH(path)

delay = GET_DELAY(path)

utilization = GET_UTILIZATION(path)

status = GET_STATUS(path)

IF status = "FAILED" THEN

score = -1

ELSE

normBandwidth = NORMALIZE_BANDWIDTH(bandwidth)

normDelay = NORMALIZE_DELAY(delay)

normUtilization = NORMALIZE_UTILIZATION(utilization)

score = (0.5 * normBandwidth) +

(0.3 * (1 - normDelay)) +

(0.2 * (1 - normUtilization))

END IF

STORE bandwidth, delay, utilization, score

IF status != "FAILED" AND utilization <= 80 THEN

IF score > bestScore THEN
alternatePath = bestPath
alternateScore = bestScore
bestPath = path
bestScore = score
ELSE IF score > alternateScore THEN
alternatePath = path
alternateScore = score
END IF
END IF
END FOR

IF bestPath = NONE THEN
DISPLAY "ALERT: No valid path for", branch
CONTINUE
END IF

IF utilization(bestPath) > 80 OR status(bestPath) = "FAILED" THEN
DISPLAY "ALERT: Primary path overloaded or failed."
IF alternatePath != NONE THEN
SELECT alternatePath
DISPLAY "Switched to alternate path:", alternatePath
ELSE
DISPLAY "No alternate path available."
END IF
ELSE
SELECT bestPath
DISPLAY "Best path for", branch, "=", bestPath
DISPLAY "Quality Score =", bestScore
END IF
END FOR

WAIT monitoringInterval
END WHILE
END

Output:

input

```
Bandwidth: 70 Mbps
Delay: 30 ms
Utilization: 75 %
Quality Score: 0.61
Best path selected.

Branch: B4
Bandwidth: 85 Mbps
Delay: 15 ms
Utilization: 85 %
Quality Score: 0.71
ALERT: Link is overloaded.
Switching to alternate path.

Branch: B5
Bandwidth: 60 Mbps
Delay: 40 ms
Utilization: 60 %
Quality Score: 0.56
Best path selected.

Branch: B6
Bandwidth: 95 Mbps
Delay: 10 ms
Utilization: 35 %
Quality Score: 0.88
Best path selected.

Monitoring completed.
```

Explanation

The algorithm does not choose a path only by distance. It evaluates bandwidth, delay and utilization together. A high-bandwidth, low-delay and lightly utilized path receives a better score. Failed or overloaded paths are avoided, and an alternate path is selected when necessary. Periodic re-evaluation allows routing decisions to adapt as network conditions change.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient